Scaling up RHD screening and prophylaxis pays off in Indonesia and Uganda, on provisional parameters

Technical review of the rheumatic heart disease investment-case model

William Garcia

University of Washington · World Bank

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We ask whether scaling up echo screening and prophylaxis is worth its cost

1. **Health:** how many rheumatic heart disease (RHD) deaths does scale-up avert in Indonesia and Uganda, 2026–2100?
2. **Services and cost:** what screening, prophylaxis and surgery does it add, at what cost?
3. **Value:** does the monetised benefit exceed the incremental cost?

Scope: national, both sexes, ages 0–100+; secondary prevention only.

About 2.2 million people live with RHD in the two countries, and few receive prophylaxis

2.16 million

people living with RHD in Indonesia and Uganda, 2023

Baseline optimal-treatment coverage in the model: 4% in Indonesia, 0.75% in Uganda.

GBD 2023 RHD prevalence, counts summed over age and sex (Indonesia 1,400,743; Uganda 762,663). Amber marks provisional assumptions throughout.

The model extends an established RHD investment case, but its stage parameters are provisional

Builds on

- Investment-case design of Coates et al. (2021): in the African Union, secondary prevention with secondary and tertiary care returned a benefit–cost ratio (BCR) of 4.7 to 2030
- World Heart Federation (WHF) echo stages A–D (Rwebembera et al., 2023)
- GBD 2023 burden; UN World Population Prospects (WPP) 2024

Boundary

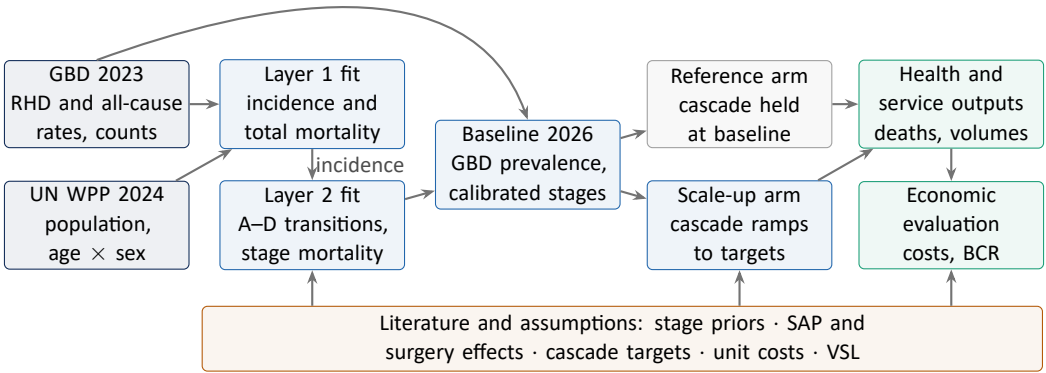
Deterministic point estimates; no sensitivity analysis yet.

Stage parameters are fitted to totals and only partly identified.

Projections are model-conditional scenarios, not forecasts.

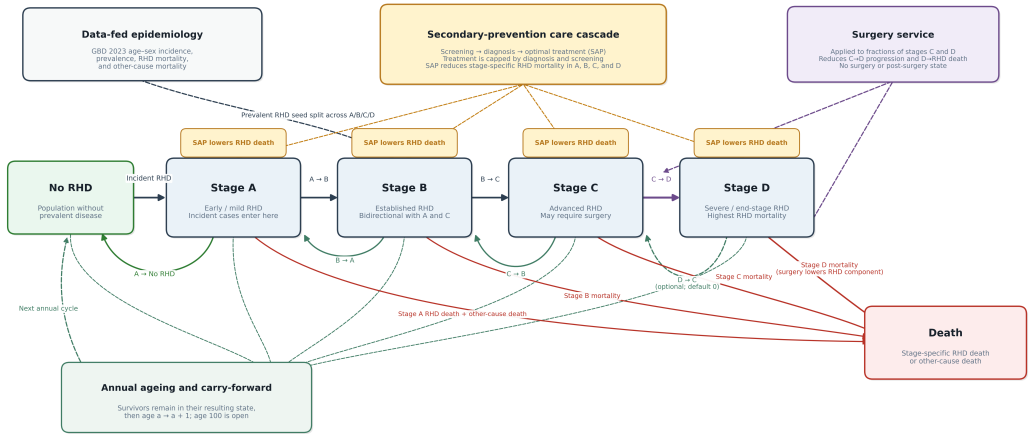
How the model works

Four linked steps turn GBD and WPP inputs into health and economic contrasts



Layer 2 and both arms run the same annual engine. SAP: secondary antibiotic prophylaxis. VSL: value of a statistical life.

Incident RHD enters stage A; SAP and surgery act only on exits



Existing project diagram (title and legend cropped). Drawn for Indonesia; the same structure runs for Uganda.

Model stages follow WHF 2023, except that stage A spans all ages and every stage can die of RHD

Stage	WHF 2023 definition	Model treatment
A	Minimal echo criteria; ages ≤ 20 only; low risk	Entry for all incident RHD, any age
B	Mild RHD; moderate or high risk	Exchanges with A and C
C	Advanced, at risk of complications	Progresses to D; regresses to B
D	Advanced with complications	Leaves only by death (D $\rightarrow$ C = 0)
RHD death	Not a stage	Allowed from A, B, C and D

Definitions from Rwebembera et al. (2023); complications include heart failure, arrhythmia, stroke or need for surgery.

Population is exogenous: WPP fixes each age–sex cell, and RHD stocks age within it

Held fixed outside the model

- Population P_t: WPP 2024, ages 0–100+, two sexes
- Background deaths: GBD all-cause minus RHD, 2023
- Incidence: calibrated 2019 rate $\times$ trend $^{t-2019}$

Evolved by the model

- Stocks A–D by age and sex
- Susceptible pool $= P_t - (A + B + C + D)$
- Annual cycles, 2026–2100
- 2026 stock: GBD prevalence $\times$ population

Incidence trend: 0.999 per year (Indonesia), 0.985 (Uganda). The 2026 stock is split by the Layer-2 stage shares for each age and sex.

Competing exits are split on the hazard scale, so every stock balances exactly

Each stage faces several annual risks at once; converting them to hazards keeps every outflow non-negative.

$$p_{s\to e} = \left(1 - e^{-H_s}\right)\frac{h_e}{H_s}, \qquad h_e = -\ln\left(1 - p_e\right), \qquad H_s = \sum_{e\in\mathcal{E}_s} h_e$$

Stage s keeps the share e^{-H_s}. $\mathcal{E}_s$ holds the exits available from s — for stage A: to no RHD, to B, RHD death, other-cause death. All probabilities are per year.

Each year draws incidence from the susceptible pool, applies exits, then ages survivors

$$A_{t+1}(a{+}1) = A_t(a)\, e^{-H_A} + B_t(a)\, p_{B\to A} + \left[P_t(a) - S_t(a)\right] \iota_t(a)$$

$A_t(a)$: stage-A persons aged a entering year t, for each sex. $S_t = A_t + B_t + C_t + D_t$. P_t: WPP population. ι_t: calibrated annual incidence probability. Stages B–D follow the same form.

Observed data anchor demography and burden; stage dynamics are calibrated around provisional priors

Input	Source	Role	Status
Population	UN WPP 2024	Age × sex × year	Observed
RHD prevalence	GBD 2023	Seeds the 2026 stock	Observed
Background deaths	GBD 2023, all-cause — RHD	Competing risk	Observed
RHD incidence	GBD 2023 × Layer 1	Inflow to stage A	Calibrated
A–D transitions	Layer 2, loose priors	Stage dynamics	Partly fitted
Stage RHD mortality	Layer 2, tight priors	RHD deaths	Provisional
Incidence trend	Assumption	Scales incidence	Provisional

Effects, costs and valuation rest on literature and provisional values

Input	Value	Role	Status
SAP effect	55% fewer RHD deaths	Multiplies RHD death	Mechanism open
Surgery effect	85% on two exits	Scaled by reach	Literature
Cascade targets	80% / 80% / 65%	Scale-up arm by 2050	Policy scenario
Cascade baselines	Country-specific	Both arms in 2026	Provisional
Unit costs	Screening, SAP, surgery	Cost per service	Undocumented
GDP per capita	2019 base, growing	Benefit valuation	Uganda unclear
VSL, discounting	30× GDP pc; 3%	Benefits and costs	Literature

Effect sizes and the VSL multiple follow Coates et al. (2021). Surgery acts on C→D and D→death. Targets: screening / diagnosis / treatment.

Calibration fits GBD incidence first, then stage dynamics inside the production engine

Layer 1: aggregate (2000–2019)

- Targets: GBD RHD incidence, prevalence, deaths
- Free: incidence and mortality multipliers per age band × sex (20 per country)
- Log-squared loss, priors, smoothness
- Incidence multipliers: 0.91–1.00

Layer 2: structural (2010–2019)

- Runs the engine from a 60-year burn-in
- Targets: GBD prevalence and RHD deaths
- Free: 6 transitions, stage-D level, 3 severity ratios, residual multiplier
- Priors: mortality log-sd 0.35; transitions logit-sd 2

Calibration undershoots GBD prevalence by about 30% and overshoots Uganda's RHD deaths 2.3×

	GBD prevalence	Model / GBD	GBD RHD deaths	Model / GBD
Indonesia	1,323,768	0.67	5,614	0.83
Uganda	658,208	0.73	360	**2.32**

Layer-2 fit in 2019, the last calibration year; counts summed over age and sex. Mortality stayed near its clinical priors (residual multiplier 1.11 Indonesia, 0.85 Uganda).

Uganda's GBD case fatality lies below the lowest plausible stage mortality, so no plausible fit matches it

Annual rate, last calibration year	Indonesia	Uganda
GBD case fatality (deaths / prevalence)	0.424%	**0.055%**
Model case fatality	0.525%	0.175%
Lowest stage mortality (stage A)	0.130%	**0.072%**

Even if every person with RHD were in stage A, Uganda's model would imply more deaths than GBD records. GBD's RHD prevalence and death estimates are hard to reconcile elsewhere too (Coates et al., 2021, appendix).

A one-at-a-time sweep flags Indonesia's C⟶B and C⟶D transitions as prior-dominated

Transition	Indonesia	Uganda
A ⟶ no RHD	0.10	0.14
A ⟶ B	0.19	0.14
B ⟶ A	0.38	0.33
B ⟶ C	0.43	0.24
C ⟶ B	**0.52**	0.43
C ⟶ D	**0.52**	0.38

Flatness: share of a parameter's range within the near-optimal objective band, others held at the optimum; amber = classified prior-dominated by the calibration. Joint identification was not tested.

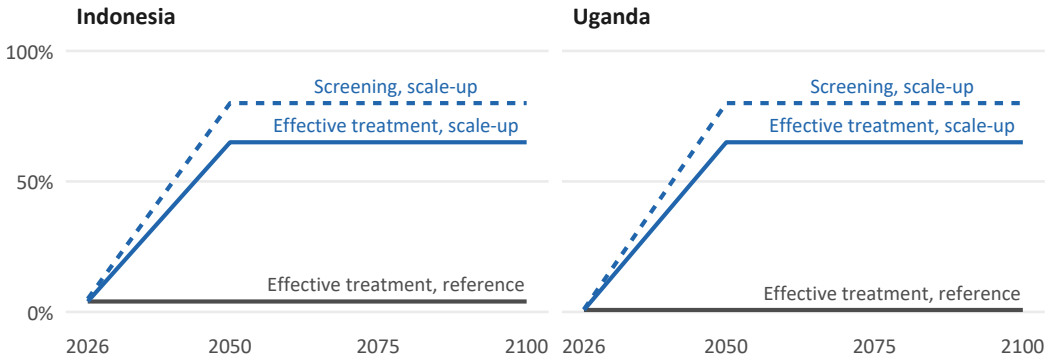
What the comparison changes

Only the care cascade differs between arms; surgery coverage is identical in both

	Indonesia		Uganda	
Cascade step (denominator)	Reference	Scale-up	Reference	Scale-up
Screening (everyone)	5%	5% → 80%	1%	1% → 80%
Diagnosis (people with RHD)	5%	5% → 80%	0.9%	0.9% → 80%
SAP (people with RHD)	4%	4% → 65%	0.75%	0.75% → 65%
Surgery (C, D in need)	5%	5%	3%	3%

Scale-up ramps linearly from 2026 to 2050, then holds to 2100. Surgery need per year: 3% of C and 20% of D (Indonesia); 10% and 40% (Uganda).

Treatment coverage reaches 65% by 2050 and then holds; screening never constrains it



Effective treatment = min(screening, diagnosis, treatment); with these targets it always equals nominal treatment coverage.

SAP scales every stage's RHD death probability down in proportion to effective coverage

$$\tilde{m}_{s,t} = m_s\left(1 - \rho\, c_t^{\text{tx}}\right)\phi_s, \qquad c_t^{\text{tx}} = \min\left(c_t^{\text{scr}},\, c_t^{\text{dx}},\, c_t^{\text{trt}}\right), \qquad \phi_D = 1 - \rho^{\text{surg}} f_D\, c^{\text{surg}}$$

m_s: calibrated stage-s RHD death probability per year. ρ = 0.55: SAP relative risk reduction. $\phi_s = 1$ for A–C. At the 65% target the multiplier is 0.64, a 34% (Indonesia) and 35% (Uganda) cut relative to the reference arm. No effect on incidence or progression.

Costs are unit prices times modelled volumes, and screening is priced on the whole population

Service	Annual volume	Indonesia	Uganda
Echo screening	Coverage × whole population	US$1.10	US$0.40
Diagnosis	Coverage × prevalent RHD	not costed	not costed
SAP (optimal treatment)	Coverage × prevalent RHD	US$110	US$40
Valve surgery	Need × coverage × C, D	US$9,000	US$5,000

Unit costs per person screened, per person-year on SAP and per surgery; sources and price year are not recorded in the pipeline. Implied perspective: programme delivery cost.

Benefits follow the full-income approach: VSL plus one year of GDP per capita for each death averted

- Benefit$_t$ = deaths averted$_t$ $\times$ (VSL$_t$ + GDP per capita$_t$), with VSL$_t$ = 30 $\times$ GDP per capita$_t$
- GDP per capita grows from 2019: US$4,150 at 3.0% (Indonesia), US$1,340 at 3.2% (Uganda)
- Coates et al. (2021) combined VSL, roughly 30 times GDP per capita, with GDP gains

Possible double count

The GDP term may overlap VSL. It is 1/31 of every benefit, so dropping it lowers both ratios by 3.2%.

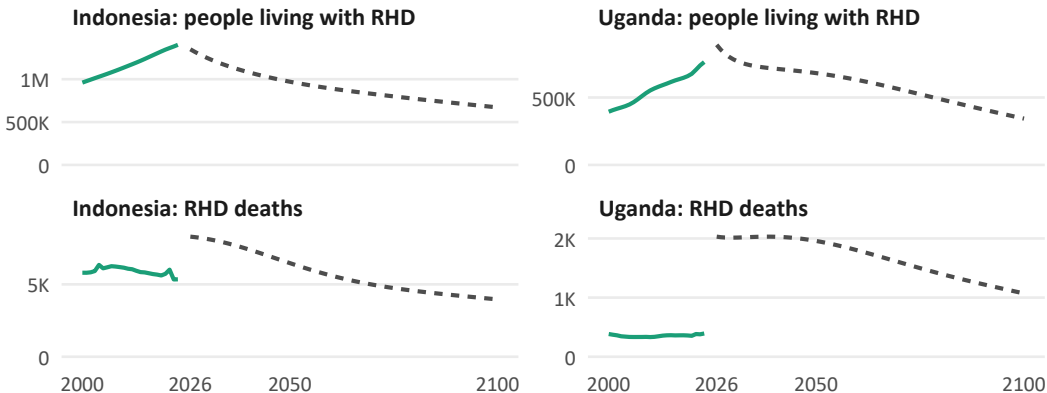
The decision metric is the discounted benefit–cost ratio over 2026–2100

$$\text{BCR} = \frac{\sum_t \delta_t \, \Delta D_t \, (\text{VSL}_t + \text{GDPpc}_t)}{\sum_t \delta_t \, \Delta C_t}, \qquad \delta_t = (1+r)^{-(t-2026)}$$

ΔD_t: RHD deaths averted (reference minus scale-up). ΔC_t: incremental cost in US$. r = 3% for costs and benefits. Net benefit = numerator − denominator. Cost per death and per disability-adjusted life year (DALY) are secondary outputs (backup); they answer a different question.

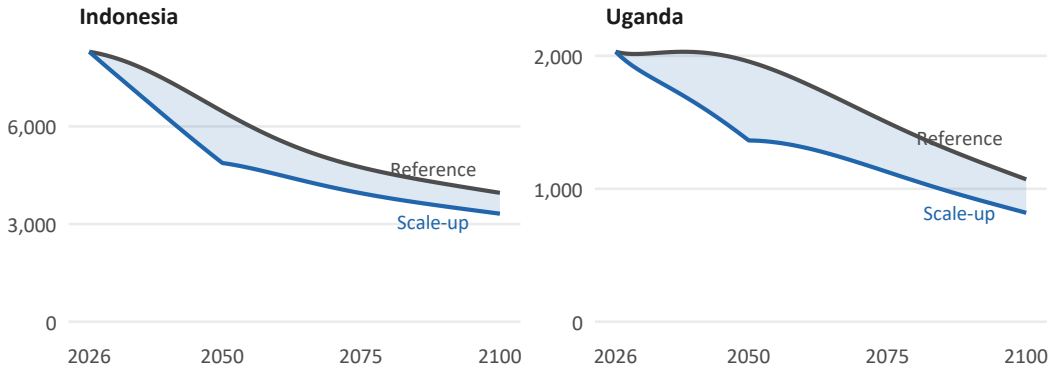
What the model projects

Projected RHD deaths start 1.6× (Indonesia) and 5.2× (Uganda) above GBD 2023



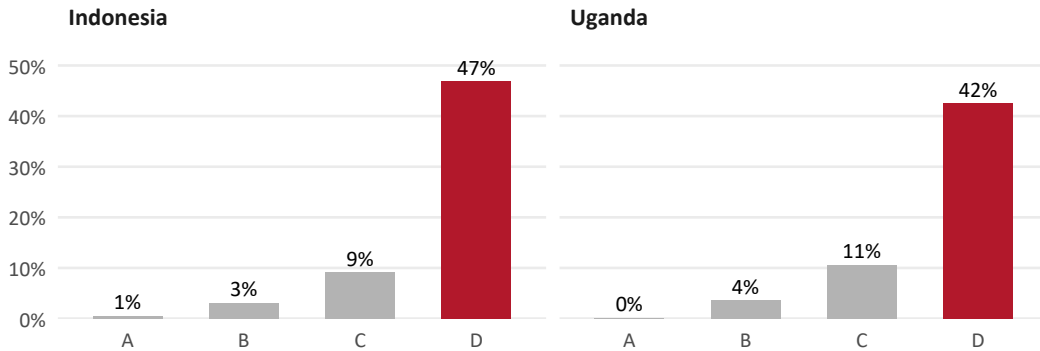
Solid green: GBD 2023, 2000–2023. Dashed grey: model reference arm, 2026–2100. All ages, both sexes.

Scale-up averts about 67,700 RHD deaths in Indonesia and 27,200 in Uganda by 2100



Annual RHD deaths; shaded = deaths averted. Cumulative: 16% (Indonesia) and 22% (Uganda) of reference-arm RHD deaths, 2026–2100.

Averted deaths become survivors: stage D is 47% (Indonesia) and 42% (Uganda) larger by 2100



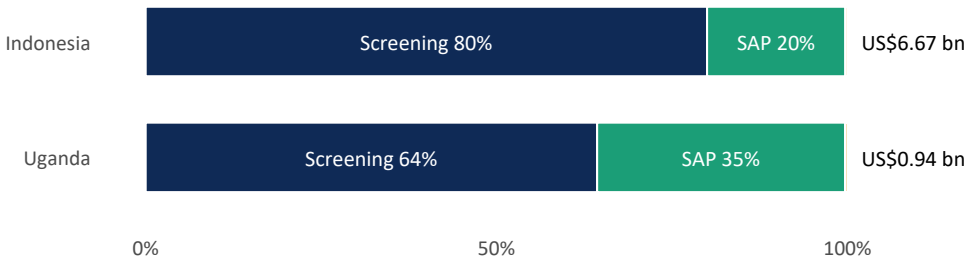
Prevalent cases, scale-up vs reference, 2100. RHD death rate in 2050: 25% (Indonesia) and 30% (Uganda) lower, against a 34%–35% cut in stage death probabilities.

By 2050 the scale-up screens 257 million people a year in Indonesia to keep 648,100 on SAP

	Indonesia		Uganda	
Per year, 2050	Reference	Scale-up	Reference	Scale-up
People screened	16.0 m	**256.8 m**	0.9 m	**68.9 m**
Prevalent RHD diagnosed	49,000	**797,700**	6,200	**552,000**
People on SAP	39,200	**648,100**	5,100	**448,500**
Valve surgeries	511	**618**	208	**236**

Counts from the model's service trace; m = million. Diagnosis and SAP volumes are shares of the prevalent RHD stock; screening is a share of the whole population.

Echo screening is 80% of Indonesia's and 64% of Uganda's discounted incremental cost



Discounted (3%, base 2026) incremental cost, 2026–2100, by component; surgery is 0.3% (Indonesia) and 0.3% (Uganda).

Benefit–cost ratios are 1.61 in Indonesia and 1.61 in Uganda over 2026–2100

Discounted to 2026 at 3%	Indonesia	Uganda
Incremental cost	US$6.67 bn	US$0.94 bn
Monetised benefit	US$10.71 bn	US$1.51 bn
Net benefit	US$4.04 bn	US$0.57 bn
Benefit–cost ratio	**1.61**	**1.61**
RHD deaths averted (undiscounted)	67,680	27,165

US$ (GDP per capita anchored to 2019; unit-cost price year undocumented). Deterministic point estimates.

Ending the horizon in 2050, as the abstract did, lowers the ratios to 1.45 and 1.28

	Indonesia		Uganda	
	2026–2050	2026–2100	2026–2050	2026–2100
RHD deaths averted	23,800	67,700	7,900	27,200
Benefit–cost ratio	**1.45**	**1.61**	**1.28**	**1.61**
Net benefit (discounted)	US$1.16 bn	US$4.04 bn	US$0.09 bn	US$0.57 bn

2026–2050 columns are deck-only sums of the saved runs to 2050; the model runs forward only, so this equals a 2050 run with the same inputs. The submitted abstract presented these runs as hypothetical countries.

What must be resolved

The engine and pipeline are implemented; the parameters that drive deaths and costs are provisional

Implemented

- A–D engine, hazard exits
- Two-layer calibration
- Two arms, both countries
- Costing and BCR

Provisional

- Stage transitions, mortality
- SAP mechanism
- Screening denominator
- Unit costs, Uganda GDP

Planned

- Stage-D-only death option
- SAP effect on progression
- Surgery pathway
- Sensitivity analyses

72% of Uganda's and 45% of Indonesia's modelled RHD deaths occur outside stage D

	Deaths outside D	Model stage-D rate	Proposed stage-D death rate	
			Literature proxy	GBD heart-failure proxy
Indonesia	45%	9.1%	2.4%	11.9%
Uganda	72%	6.6%	0.3%	6.0%

Reference arm, 2026. Proposed rates divide GBD 2023 RHD deaths by an estimated stage-D count (a literature stage-D share, or GBD RHD heart-failure prevalence); collaborator worksheet, October 2026.

Calibrated C→B regression sits below every published estimate, and D→C is fixed at zero

Annual probability	Literature range (studies)	Indonesia	Uganda
A → no RHD	0.007–0.249 (11)	0.071	0.178
A → B	0.002–0.072 (15)	0.011	0.008
B → A	0.004–0.203 (11)	0.016	0.006
B → C	0.001–0.125 (9)	0.044	0.013
C → B	0.012–0.081 (4)	**0.005**	**0.005**
C → D	0.006–0.250 (4)	0.082	0.036
D → C	0.004–0.071 (3)	**0.000**	**0.000**

Annualised ranges from the collaborators' September 2026 literature table; amber bold = outside the range. Ranges pool cohorts with and without SAP.

The 55% effect is applied to mortality, but its sources report effects on progression

In the model

- 55% lower RHD death risk in every stage, scaled by coverage
- No effect on incidence or progression

In the sources

- Coates et al. (2021): 55% (7–78%) on progression to severe RHD
- WHF 2023: in a trial in children with stage A or B, SAP slowed progression; number needed to treat 13 (Rwebembera et al., 2023)

A mortality effect acts at once; a progression effect is delayed and works by preventing stage D.

The strongest objection: totals do not identify stages, and stages drive the benefit

A skeptic would say

GBD gives only total prevalence and deaths. Many stage splits fit equally well, yet the SAP benefit depends on how many people sit in high-mortality stages.

What the evidence supports

Results are conditional on one calibrated equilibrium. Stage-level echo targets would pin the split; until then, ranges should come from near-optimal sets and sensitivity analysis.

Remaining limitations bias results in directions we can name

Limitation	Likely effect on results	What resolves it
Mortality attribution	Uganda deaths averted high	Stage-D option; local deaths
Transferability	Direction unknown	Country cohort or registry data
Implementation	Effect per dollar high	Adherence and retention data
Screening denominator	Cost high, BCR low	Target ages, screening interval
Valuation	GDP term adds 3.2%	VSL and discount sensitivity
Uncertainty	Spread unknown	Probabilistic, one-way analyses

Five updates, in this order, should precede any investment conclusion

1. **Mortality structure and stage-D rate** — affects deaths, stage mix, every result
2. **Transitions from the literature table** (D→C > 0, C→B in range) — stock and deaths
3. **SAP mechanism:** mortality, progression or both — size and timing of benefit
4. **Screening denominator, interval and unit-cost sources** — 64%–80% of incremental cost
5. **Echo stage targets and sensitivity analyses** — confidence in all of the above

On current saved runs, scale-up returns about US$1.6 per dollar in both countries through 2100—but stage mortality, the SAP mechanism and the screening denominator must be settled first.

Five collaborator decisions will define the next model version

1. Should RHD deaths occur only in stage D, and which stage-D rate should anchor it?
2. Does SAP slow progression, cut mortality, or both, and in which stages?
3. Which published transitions replace the priors; should D$\rightarrow$C regression be allowed?
4. What target ages and screening interval should define the cost denominator?
5. Does surgery return people to stage C or only lower death, and does it apply to stage C?

Backup slides hold detail for questions and sit outside the 40-minute budget

- Cycle order and full engine equations
- Calibration objectives, parameters, multipliers, fit
- Near-optimal sets; literature transitions
- Economic parameters and detailed outputs
- Budget impact; abstract reconciliation
- Sensitivity status; discrepancies; provenance

Within each annual cycle, the engine runs seven steps in a fixed order

1. Stocks A–D enter year t already aged; $S_t = A_t + B_t + C_t + D_t$
2. Susceptible pool $W_t = \max(P_t - S_t, 0)$; incident cases $N_t = W_t\,\iota_t$
3. Coverage sets the SAP multiplier and surgery reach $f_C c^{\text{surg}}$, $f_D c^{\text{surg}}$
4. Each stage splits its exits on the hazard scale
5. Year-end stocks collect stays and inflows; N_t joins A
6. Record deaths and volumes: screened $= c^{\text{scr}} P_t$; diagnosed, on SAP $= c\,S_t$; surgeries $= f\,c^{\text{surg}} \times$ C, D
7. Survivors age one year; age 0 starts empty; 100+ is open

The four stage updates share one form; only stage D lacks an exit to a higher stage

$$
\begin{aligned}
A_t^{+} &= A_t\, e^{-H_A} + B_t\, p_{B\to A} + W_t\, \iota_t \\
B_t^{+} &= B_t\, e^{-H_B} + A_t\, p_{A\to B} + C_t\, p_{C\to B} \\
C_t^{+} &= C_t\, e^{-H_C} + B_t\, p_{B\to C} + D_t\, p_{D\to C} \\
D_t^{+} &= D_t\, e^{-H_D} + C_t\, p_{C\to D}(1 - \rho_{CD}^{\text{surg}} f_C c^{\text{surg}})
\end{aligned}
$$

RHD deaths $= \sum_s s_t\, p_{s\to\text{RHD}}$; all-cause deaths $=$ RHD deaths $+ \mu P_t$, with μ the background mortality. Each p is the hazard-split outflow; $A_{t+1} = \text{shift}(A_t^{+})$.

Layer 1 minimises a weighted log error with priors and smoothness on band multipliers

$$L_1 = \sum_{X \in \{I,P,D\}} w_X \frac{\sum_j \omega_j \left[\ln(X_j^{\text{mod}} + \varepsilon) - \ln(X_j^{\text{GBD}} + \varepsilon) \right]^2}{\sum_j \omega_j} + \lambda \sum_g \left[\frac{\eta_g^2}{\sigma_\alpha^2} + \frac{\zeta_g^2}{\sigma_\beta^2} \right] + \lambda_s \sum_g \left[(\Delta^2 \eta_g)^2 + (\Delta^2 \zeta_g)^2 \right]$$

$\eta_g = \ln \alpha_g$ (incidence), $\zeta_g = \ln \beta_g$ (mortality), bands 0-14, 15-24, 25-44, 45-64, 65+. Weights w = 1, 1, 2; ω inverse-variance from GBD uncertainty intervals (used). σ_α = 0.75, σ_β = 0.5. Optimiser: multistart L-BFGS-B, 8 random starts plus baseline.

Layer 2 orders stage mortality by construction and penalises departures from priors

$$m_s = r_s\, m_D \;(s = A, B, C), \qquad r_A \le r_B \le r_C < 1$$

$$L_2 = \sum_{\text{sex}} \left(w_P \bar{\ell}_P + w_D \bar{\ell}_D\right) + \sum_s \frac{(\ln m_s - \ln m_s^0)^2}{\sigma_m^2} + \sum_k \frac{(\operatorname{logit} p_k - \operatorname{logit} p_k^0)^2}{\sigma_p^2} + \frac{(\ln \kappa)^2}{\sigma_\kappa^2}$$

$\bar{\ell}$: mean squared log error over age groups and years 2010–2019; w_P, w_D = 1, 2. κ: residual mortality multiplier in [0.5, 2], σ_κ = 0.25. $m_D \in$ [0.03, 0.25]. σ_m = 0.35, σ_p = 2. The 2026 stage shares are the equilibrium after a 60-year burn-in.

Calibrated transitions move away from the priors; stage mortality stays near them

Annual probability	Prior centre	Indonesia	Uganda
A → no RHD	0.050	0.0713	0.1777
A → B	0.020	0.0111	0.0083
B → A	0.010	0.0161	0.0058
B → C	0.030	0.0437	0.0134
C → B	0.005	0.0050	0.0047
C → D	0.060	0.0818	0.0362
D → C	0.000	0.0000	0.0000
RHD death, stage A	0.0010	0.0013	0.0007
RHD death, stage B	0.0050	0.0057	0.0039
RHD death, stage C	0.0200	0.0226	0.0164
RHD death, stage D	0.0800	0.0906	0.0663
Residual mortality multiplier	1	1.114	0.851
Equilibrium stage-D share	derived	0.030	0.007

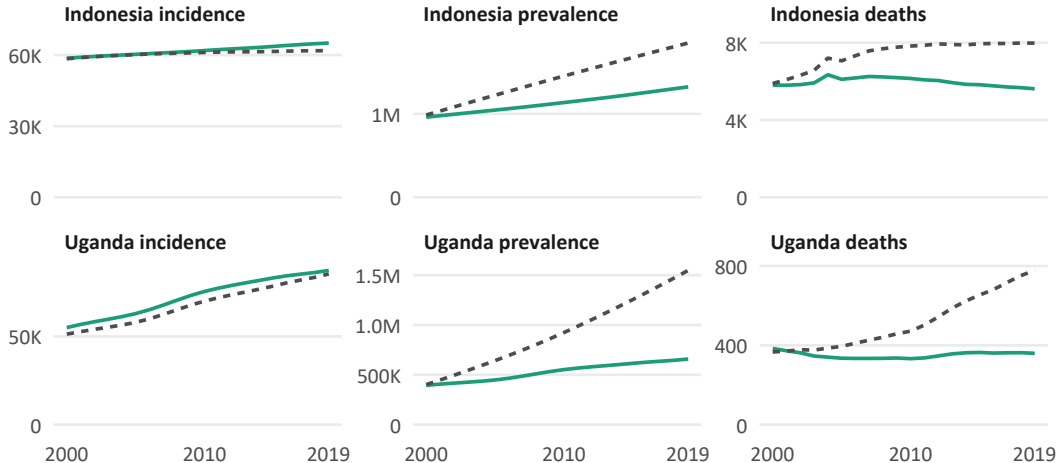
Stage mortality shown is the effective value used in projection (untreated estimate × residual multiplier). D→C fixed, not estimated.

Layer-1 multipliers stay close to one, so calibrated incidence follows the GBD pattern

	Indonesia		Uganda	
Age band	Incidence α	Mortality β	Incidence α	Mortality β
0-14	0.98 / 0.97	0.99 / 0.99	0.92 / 0.91	0.97 / 0.97
15-24	0.98 / 0.98	0.99 / 0.99	0.95 / 0.94	0.98 / 0.97
25-44	0.99 / 0.98	0.99 / 0.98	0.96 / 0.96	0.97 / 0.96
45-64	1.00 / 0.99	0.98 / 0.98	0.98 / 0.98	0.96 / 0.96
65+	1.00 / 1.00	0.96 / 0.97	1.00 / 1.00	0.96 / 0.98

Female / male multipliers on GBD incidence (α) and case fatality (β), 2000–2019. Only α feeds the projection. The penalised objective improved on its baseline for every sex.

Layer 1 matches GBD incidence but cannot reconcile it with GBD prevalence



Green: GBD 2023 target counts. Dashed grey: Layer-1 aggregate model. Summed over age groups and sex.

The near-optimal sets are tight, but only 2 random starts were run

Country	Objective	m_D	Residual	A→no RHD	B→C	C→D
Indonesia	31.111	0.0814	1.114	0.0713	0.0437	0.0818
Indonesia	31.111	0.0814	1.113	0.0714	0.0433	0.0819
Indonesia	31.111	0.0814	1.114	0.0712	0.0438	0.0820
Uganda	25.821	0.0780	0.851	0.1777	0.0134	0.0362
Uganda	25.821	0.0780	0.851	0.1775	0.0134	0.0361
Uganda	25.821	0.0779	0.851	0.1767	0.0134	0.0362

Prior-centre objective: 34.0 (Indonesia), 46.9 (Uganda); minimum 31.1 and 25.8. Near-optimal band set by `STRUCT_NEAROPT_DELTA` in the calibration configuration.

Published annual transition probabilities vary 7- to 108-fold across studies

Transition	Studies	Minimum	Maximum	Max / min	Indonesia	Uganda
A → no RHD	11	0.0074	0.2490	34	0.0713	0.1777
A → B	15	0.0015	0.0723	48	0.0111	0.0083
B → A	11	0.0042	0.2025	48	0.0161	0.0058
B → C	9	0.0012	0.1253	108	0.0437	0.0134
C → B	4	0.0119	0.0810	7	0.0050	0.0047
C → D	4	0.0057	0.2497	44	0.0818	0.0362
D → C	3	0.0042	0.0710	17	0.0000	0.0000

Table 2 of the collaborators' September 2026 worksheet: each study's probability annualised as $1 - (1 - p)^{1/\text{years}}$. Includes both arms of one SAP trial (control and perfect adherence).

Economic parameters differ by country in GDP, growth and unit costs

Parameter	Indonesia	Uganda
Discount rate	3%	3%
Discount base year	2026	2026
GDP per capita (base year)	US$4,150	US$1,340
GDP per capita base year	2019	2019
Real GDP per capita growth	3.0%	3.2%
VSL multiple of GDP per capita	30	30
DALYs per RHD death	30	30
Cost per person screened	US$1.10	US$0.40
Cost per person-year on SAP	US$110	US$40
Cost per surgery	US$9,000	US$5,000

As saved with the economic results. DALY: disability-adjusted life year (deaths only, no years lived with disability).

Cost-effectiveness outputs exist but mix discounted and undiscounted terms

2026–2100	Indonesia	Uganda
Cost per death averted	US$98,490	US$34,587
Cost per DALY averted	US$8,051	US$3,023
BCR without GDP term (deck-only)	1.55	1.56
Mean annual incremental cost per person (deck-only)	US$0.85	US$0.39

Cost per death: discounted cost / undiscounted deaths averted. Cost per DALY: discounted cost / (discounted deaths averted × 30). Per-person cost: undiscounted incremental cost / population, averaged over years.

Annual incremental cost levels off after the 2050 ramp in both countries

	Indonesia, US$ m			Uganda, US$ m		
Year	Reference	Scale-up	Increment	Reference	Scale-up	Increment
2030	27.6 m	82.6 m	55.1 m	1.4 m	8.0 m	6.5 m
2040	27.3 m	221.2 m	193.9 m	1.6 m	25.8 m	24.2 m
2050	26.6 m	359.3 m	332.7 m	1.6 m	46.7 m	45.1 m
2060	25.6 m	354.8 m	329.2 m	1.5 m	49.3 m	47.7 m
2070	24.7 m	347.5 m	322.8 m	1.4 m	50.6 m	49.2 m
2080	23.8 m	338.0 m	314.2 m	1.3 m	50.9 m	49.5 m
2090	22.9 m	326.4 m	303.5 m	1.3 m	50.2 m	48.9 m
2100	21.9 m	313.1 m	291.2 m	1.2 m	48.7 m	47.5 m

Undiscounted annual programme cost by arm (screening + SAP + surgery), selected years.

Truncated saved runs reproduce the abstract's Uganda figures; Indonesia's differ

	Uganda (low-income illustration)		Indonesia (upper-middle-income)	
	Abstract	Saved run	Abstract	Saved run
RHD deaths averted by 2050	7,900	7,900	23,000	23,800
Death-rate reduction in 2050	30%	30%	24%	25%
Benefit–cost ratio	1.28	1.28	1.42	1.45

Abstract values parsed from the submitted WCRHD 2026 abstract, which presented hypothetical countries. Saved-run values are sums to 2050 of the current saved outputs.

No sensitivity analysis exists yet; these analyses are planned

- **Probabilistic:** sample stage transitions and mortality from near-optimal sets and prior ranges
- **One-way:** SAP effect across 7–78% (the range in Coates et al., 2021); VSL multiple; discount rate; screening denominator; unit costs
- **Configuration:** Indonesia incidence trend 0.999 (saved run) against the current configuration value
- **Structural:** stage-D-only mortality; SAP on progression; surgery returning patients to C

The pipeline is deterministic; none of the results in this deck carry uncertainty intervals.

Six discrepancies between documents, configuration and code are logged

Item	Detail	Action
Indonesia incidence trend	Saved run 0.999; scripts/00_run_all.R now 0.985	Re-run Indonesia
Uganda GDP per capita	Active US$1,340; superseded comment cites about US$822 (World Bank 2019)	Confirm source
State diagram	Subtitle names Indonesia only; other-cause death drawn from stage A only	Redraw labels
Seed stage split	Report text cites a literature split; 05 seeds Layer-2 equilibrium shares	Update report
Layer-1 mortality anchor	Documented as anchoring Layer 2; 04b uses only Layer-1 incidence	Update docs
Diagnosis cost	Volumes tracked in 06; no unit cost in 08	Decide

Provenance (1/2): calibration, validation and health results trace to saved objects

Result	Script	Artifact	Object	Shape	Transform
Fit, case fatality	04b	data/<C>/calibrated_rhd_parameters.rds	$stage_calibration$excess_death_attribution	scalars; persons, deaths per year	ratios; deaths / prevalence
Identifiability	04b	same	$stage_calibration$identifiability	11 × 3	none
Validation	01, 06	data/<C>/temp_baseline_rates_gbd.rds; output/<C>/out_model/<C>.rds	GBD Number; $wsd sick, dead (ref)	age × sex × year; persons	sum over age and sex
Deaths averted	06	output/<C>/out_model/<C>.rds	wsddead	101 ages × 2 sexes × 75 years × 2 arms	ref − sap; cumulative; ≤ 2050
Stage stocks, deaths	06	same	$stages A–D, rhd_deaths_A–D	same; persons	sums; percent difference

Provenance (2/2): service, cost and economic results trace to saved objects

Result	Script	Artifact	Object	Shape	Transform
Coverage, scenarios	05	data/<C>/baseline_state.rds	$states[[1]]$coverage	vectors of 75 years	none
Service volumes	06	output/<C>/out_model/<C>.rds	$stages n_screened, n_diagnosed, n_on_optimal_treatment, total_surgeries	persons or surgeries per year	sum by arm and year
Cost composition	06, 08	same + rhd_economic_results.rds	$stages volumes × $params unit costs	US$	× discount factor; sap — ref
Economic summary	08	output/<C>/tables/rhd_economic_results.rds	$summary	1 row per location	none
Horizon	08	same	$incremental	75 years	sums to 2050
Literature, stage-D proxies	none	dev/*.xlsx	Table 2; All-ages rows	probabilities	min, max; read as saved
Abstract values	none	dev/Abstract WCRHD26 DW_NM.docx	Results paragraph	text	regular expressions

In paths, <C> is Indonesia or Uganda. All paths relative to the repository root; the deck reads these files and writes nothing.

Rendering needs only the saved artifacts, R and a XeLaTeX toolchain

- Render: `rmarkdown::render("reports/slideck-technical.Rmd")` with XeLaTeX and Calibri; set `RHD_REPO_ROOT` when rendering a copy outside the repository
- Inputs: artifacts of scripts 01–08 for both countries; a missing file stops the render and names its producing script
- Software: R version 4.5.1 (2025-06-13 ucrt); rmarkdown 2.29; knitr 1.50; data.table 1.17.8; ggplot2 4.0.3; kableExtra 1.4.0
- Model outputs saved: Indonesia 2026-07-23, Uganda 2026-07-23
- Citations: `library/references.bib`, APA 7 via `reports/apa.csl`

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